KAPASA MAKASA UNIVERSTY

CYS 339: FRAUD AND DETECTION TECHNOLOGIES

Group Lab Assignment

1. Background & Report (Written Submission)

a) **Introduction**: Explain the assigned technique (e.g., Gap Analysis) and its relevance to fraud prevention and detection.
b) **Dataset Description**: Source, attributes, and why it was selected. You are free to use your own dataset or any publicly available one.
c) **Methodology**: Tools used (Python, SQL, Excel, Power BI, etc.) and steps taken.
d) **Findings**: Key insights, visualizations, and potential fraud indicators.
e) **Conclusion**: Lessons learned and recommendations.
f) **Video (3-5 min)**: Screen recording of the analysis (e.g., Python code execution, dashboard demo etc).

The assignment is due last weekend, before our lectures begin. The submission platform will be communicated in due course.

2. Presentation/Demo (To be done during our contacts)

a) **Slides**: Overview of methodology, findings, and challenges.
b) **Live Demo**: Show the working solution (e.g., running a Python script, filtering fraud cases in Power BI, etc).
c) **Q&A**: Address questions from the audience.

PERFORMING DATA ANALYSIS TASKS

Gap Analysis (This task will be done by the lecturer)

Objective: Identify discrepancies between expected and actual behavior (e.g., transaction amounts, login times).

Deliverables:

a) A report explaining gap analysis and its fraud detection applications.
b) Code/script (Python, R, or SQL) calculating gaps and flagging anomalies.
c) Visualizations (e.g., bar charts of gaps, time-series plots with thresholds).

Example: Compare expected vs. actual sales transactions to detect inflated invoices.

Group 1: Duplicate Analysis

Objective: Detect exact duplicates (e.g., repeated transactions, cloned records).

Deliverables:

a) A methodology for identifying duplicates (e.g., hashing, SQL GROUPBY).
b) A list of duplicate records and their risk score.

Example: Find duplicate insurance claims with identical amounts, dates, and providers.

Group 2: Fuzzy Duplicate Detection

Objective: Find near-duplicates (e.g., slightly altered names, addresses).

Deliverables:

a) Code using fuzzy matching (e.g., Python's fuzzywuzzy, Levenshtein distance).
b) A scored list of potential matches (e.g., "John Doe" vs. "Jon Dowe").

Example: Detect fake accounts with similar but not identical personal details.

Group 3: Orphan Transactions

Objective: Identify transactions without matching records (e.g., payments without invoices).
Deliverables:

a) SQL/Python script to detect orphans (e.g., LEFT JOIN with NULL checks).
b) Analysis of orphaned records (e.g., total value, frequency).
   Example: Flag unauthorized payroll payments missing employee records.

Group 4: Timing Analysis

Objective: Detect unusual timing (e.g., off-hours logins, rapid-fire transactions).
Deliverables:

a) Time-based anomaly detection (e.g., Python's statsmodels for seasonality).
b) Visualizations (e.g., heatmaps of login times).
   Example: Identify employees accessing systems at 3 AM.

Group 5: Limit and Reasonableness Checks

Objective: Flag implausible values (e.g., salaries exceeding $1M, negative quantities).
Deliverables:

a) Business rules for "reasonable" thresholds.
b) A report listing violations (e.g., "Employee salary exceeds industry max").
   Example: Detect fraudulent expense claims (e.g., $10,000 "team lunch").

Group 6: Trend and Outlier Detection

Objective: Spot deviations from historical trends (e.g., sudden spikes in refunds).
Deliverables:

a) Time-series forecasting (e.g., moving averages, ARIMA).
b) Annotated outliers (e.g., "Refund requests increased by 500% in March").
   Example: Identify money laundering via unusual cash deposits.

Group 7: Unwanted Matches

Objective: Detect prohibited relationships (e.g., vendors with employee names).
Deliverables:

a) Cross-matching script (e.g., joining employee and vendor tables).
b) A risk assessment of matches (e.g., "Vendor 'Jane Smith' matches HR records").
   Example: Find conflicts of interest in procurement data.

Group 8: Aging, Stratification, Sampling

Objective: Analyze aging (e.g., overdue invoices) and stratify risk.
Deliverables:

a) Aging reports (e.g., "60% of claims are >90 days old").
b) Stratified sampling for audits (e.g., high-risk vs. low-risk groups).
   Example: Prioritize fraud investigations based on invoice age.

PERFORMING DATA PROFILING TASKS

Group 9: Frequency Tests

Objective: Check for unusual frequencies (e.g., one product in 80% of transactions).

Deliverables:

a) Histograms/top-frequency lists.

b) Risk assessment (e.g., "Product X is suspiciously dominant").

Example: Detect fake reviews from users posting 100+ times/day.

Group 10: Intensity Tests

Objective: Measure activity concentration (e.g., 90% of logins from one IP).

Deliverables:

a) Intensity heatmaps (e.g., by IP, user, region).

b) Summary of high-intensity clusters.

Example: Identify DDoS attacks or botnet activity.

Group 11: High-Risk Tests

Objective: Flag high-risk attributes (e.g., offshore accounts, shell companies).

Deliverables:

a) A risk-scoring model (e.g., "Country = Tax Haven → +10 risk").

b) A ranked list of high-risk entities.

Example: Detect politically exposed persons (PEPs) in banking data.

Group 12: Relative Size Tests

Objective: Compare item sizes to benchmarks (e.g., transactions > 3x average).

Deliverables:

a) Z-score/percentile analysis.
b) A report on "unusually large" items.

Example: Flag Ponzi scheme payouts (e.g., 10x typical returns).

Group 13: Benford's Law (Digit Frequency)

Objective: Detect unnatural digit distributions (common in fraud).

Deliverables:

a) Python implementation of Benford's Law.
b) Chi-square test results comparing actual vs. expected digits.

Example: Catch manipulated financial statements.

Group 14: GEL Concentration Test

Objective: Identify excessive concentration (e.g., one vendor getting 95% of contracts).

Deliverables:

a) Gini coefficient or Herfindahl index calculation.
b) A concentration risk report.

Example: Uncover bid-rigging in government contracts.

Final Deliverables (Zipped Folder)

1. **Report** (PDF/docx).
2. **Code/Scripts** (Jupyter notebook, SQL queries, Excel files etc).
3. **Video Walkthrough** (MP4).
4. **Presentation Slides** (PPT/PDF).
5. **Dataset** (if not publicly available).

Suggested Tools (You can use any other tool that suits your task; the ones listed below are merely suggestions)

a) **Python**: pandas, numpy, matplotlb, scipy, fuzzywuzzy.

b) **SQL**: Joins, window functions, CTEs.

c) **Excel**: PivotTables, conditional formatting.

d) **Power BI/Tableau**: Dashboards for visualization.